

Peace Umolu

Electronics & Embedded Systems Engineer

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PROFESSIONAL SUMMARY

Electronics & Embedded Systems Engineering student (3rd Year, First-Class Honors) with experience developing bare-metal firmware for ARM Cortex-M microcontrollers. Built BMS test rig with FreeRTOS, 16-bit ADC with DMA, I2C/SPI drivers, and Python automated testing. Currently building embedded systems portfolio with OrCAD schematics and Quarto documentation. Maintains First-Class standing while working 36+ hours weekly.

TECHNICAL SKILLS

Languages	Embedded C, C++, Python (PyVISA, PySerial, Pytest), Java, Bash
Microcontrollers	ARM Cortex-M33 (ADuCM410), Cortex-M4, STM32, ESP32
Protocols	I2C, SPI, CAN Bus, UART, MDIO
Tools	Keil MDK, IAR EWARM, OrCAD, KiCad, Git, Linux, Quarto
Hardware	FreeRTOS, HAL, Register-Level Programming, DMA, ECC
Lab	DSO, Logic Analyzers, Multimeters, Function Generators

EDUCATION

TUS	<i>Limerick, Ireland</i>
– B.Eng. (Hons) Electronic Engineering with Computer Systems	<i>2024–Present</i>
– First-Class Honors. Achieved while managing 36+ hours/week employment.	

IES Cabanyal	<i>Valencia, Spain</i>
– Advanced Certificate (NFQ Level 6)	<i>2013–2015</i>
– Industrial electrical installations, PLC programming, schematics.	

TECHNICAL PROJECTS

Embedded Systems Portfolio (In Progress)	<i>2026</i>
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Developing portfolio on ARM Cortex-M platforms.

Completed: GPIO interrupts, UART CLI with Python automation, PWM motor control, 16-bit ADC with oversampling, I2C/SPI with DMA, power management with sleep modes.

In Progress: Flash bootloader with fail-safe update, FreeRTOS with tasks/queues/semaphores, OrCAD schematic and PCB layout, Quarto documentation.

Smart Single-Cell BMS & HIL Test Rig	<i>2025–2026</i>
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- Multi-threaded FreeRTOS application monitoring Vcell, Iload, temperature. 16-bit ADC with DMA circular buffering for continuous sampling.
- Non-blocking I2C and SPI telemetry drivers. Coulomb Counting SoC algorithm calibrated against OCV lookup tables.
- PyVISA test suite using pytest and pyserial to sweep voltages, log telemetry, validate fault trip within 50 ms.
- Complete system schematic and PCB layout in OrCAD.

PROFESSIONAL EXPERIENCE

Acquired Brain Injury (ABI) Ireland	<i>Tipperary, Ireland</i>
– Rehabilitation Healthcare Assistant	<i>July 2026–Present</i>
– Person-centered rehabilitation support. Maintain health, safety, and compliance documentation.	

Abbott Vascular	<i>Clonmel, Ireland</i>
– Technical Operative / Assembler	<i>March 2025–Present</i>
– Precision assembly of medical devices in high-compliance cleanroom. Adhere to SOPs.	

Access Healthcare	<i>Ireland</i>
– Healthcare Assistant (Agency)	<i>2024–Present</i>

- High-demand care across facilities. Manage 36 hours weekly alongside studies.

Inforlandia*Valencia, Spain***– IT Support Technician & Frontend Developer***2017–2021*

- Diagnosed hardware and network faults. Configured software, drivers, and UIs.

B-Market / Jobelsa*Spain***– Electronic Technician & PLC Programmer***2015–2017*

- Assembled electronic tools. Programmed PLCs for industrial production lines.

ADDITIONAL INFORMATION

Languages	Fluent in English and Spanish
Certifications	PHECC First Aid, Manual Handling, Safeguarding Adults
Driving License	Full Clean B Irish Driving License